

Omni-Scan2BIM: A ready-to-use Scan2BIM approach based on vision foundation models for MEP scenes

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ABSTRACT

Mechanical, electrical, and plumbing (MEP) systems play a crucial role in providing various services and creating comfortable environments for urban residents. In order to enhance the management efficiency of these highly complex MEP systems, as-built building information models (BIMs) are being increasingly adopted worldwide. As-built BIMs accurately represent the actual conditions of facilities, making as-built BIM reconstruction significantly important for construction progress tracking, quality assurance, subsequent facility management, and renewal. To create as-built BIMs for MEP systems, laser scanners are widely utilized to capture high-resolution images and dense 3D measurements of the environment in a fast and highly accurate manner. Despite research efforts to automatically achieve “Scan-to-BIM,” there are still gaps in applying current solutions to real-world scenarios. One of the major challenges are the limited generalization of existing methods to unseen scenarios without proper training or fine-tuning on custom-designed datasets. To address this issue, this study introduces Omni-Scan2BIM, a novel approach powered by large-scale pre-trained vision foundation models. Omni-Scan2BIM enables the recognition of MEP-related components with a single shot by integrating an all-purpose feature extraction model and a class-agnostic segmentation model. Firstly, given only a single image with a reference mask, the visual features are extracted for both the target component and the collected on-site images using the vision foundation model DINOv2. Secondly, through comparing pixel features, similarity maps are generated for the on-site images. The prior points for the class-agnostic segmentation model are sampled from the local maxima of the similarity map. Thirdly, Segment Anything Model (SAM) is leveraged to sequentially segment the target component. Finally, the target component is segmented out in 3D space based on the transformation matrix describing the spatial relationship between the 2D images and the 3D point clouds. Shape analysis and label fusion are conducted for as-built BIM modeling purpose. To validate the feasibility of the proposed technique, experiments were conducted using data collected from a real construction site in Hong Kong. The results demonstrate that the proposed Omni-Scan2BIM approach can easily generalize to unseen components with significantly improved accuracy and efficiency.

1. Introduction

The adoption of Building Information Model (BIM) has demonstrated significant potential in improving project efficiency throughout the entire lifecycle of facilities, encompassing planning, design, construction, and operation and maintenance (O&M) [1]. Among these stages, the O&M phase consumes the most time and incurs the highest costs [2]. According to the National Institute of Standards and Technology (NIST), inadequate interoperability in the Architecture, Engineering, and

Construction (AEC) industry results in an annual additional cost of \$15.8 billion, with \$10.6 billion attributed to the O&M phase [3]. The implementation of BIMs in facility management is anticipated to alleviate unnecessary expenses by enhancing communication and streamlining facility management activities. As a comprehensive information platform, BIMs provide virtual environments that enable parametric simulation, assessment of facility conditions, and identification of potential issues. These capabilities allow for the establishment of appropriate maintenance schedules to prevent severe deterioration [4].

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Moreover, BIMs offer detailed component information, including types, connections, and adjacency relationships with other components. This information assists technical staff in swiftly locating problematic areas and executing repairs efficiently, particularly when dealing with complex systems.

Mechanical, electrical and plumbing (MEP) system provides essential services and creates comfortable environments to occupants, hence improving the efficiency of the MEP system is a key objective in facility management [2]. In large-scale public projects, such as healthcare centers or biotech industries, the investment associated with MEP systems can account for over 50% of the total project cost [5]. Furthermore, MEP systems play a critical role in various civil infrastructures, including petroleum refineries and wastewater treatment facilities. Extensive research has been conducted on the implementation of BIM technology for MEP system management across different phases, ranging from design [5–7], construction [5,8,9] to O&M [2,10]. During the O&M phase, BIM technology enables more intelligent facility management by digitizing and integrating a vast amount of documents and drawings within BIMs [2]. Specifically, the digital model can be utilized to analyze the conditions of MEP systems in virtual environments, monitor component conditions through wireless sensors, swiftly identify problematic components, and prioritize routine maintenance tasks accordingly [11]. By leveraging BIM-aided O&M management, the lifespan of MEP systems can be extended, ensuring the provision of high-quality services.

In real construction projects, unlike disciplines such as architecture and structure, the disparities between the as-designed BIMs and as-built BIMs for MEP systems are substantial. These discrepancies arise from frequent design changes or deviations during assembly, rendering it impractical to directly utilize the as-designed BIMs for tracking construction progress and managing facilities during the operational stage. Consequently, BIM modelers are compelled to manually create as-built BIMs in modeling software, relying on collected as-built data such as on-site photos and point clouds. However, this method entails a heavy workload, necessitating the involvement of numerous experienced modelers for extended periods of time. Therefore, there is a pressing need to develop a reliable and efficient approach for generating accurate and updated as-built BIMs with comprehensive facility information.

To comprehensively capture the site's details, operators utilize terrestrial laser scanners (TLS), also known as Light Detection and Ranging (LiDAR), to acquire on-site photos and collect point cloud data with millimeter accuracy. This approach captures the external surfaces of objects in both image and point cloud formats [12]. Although various deep learning methods have demonstrated potential in extracting semantic information from point clouds, their applicability to unseen scenes is hindered by their reliance on purpose-built datasets. However, in parallel with the rapid advancement of large language models (LLMs) like GPT [13], significant progress has been made in the development of large-scale vision models. The vision foundation models exhibit exceptional transfer capabilities, demonstrating their ability to perform effectively on tasks and data distributions that lie outside the boundaries of their original training scope. The impressive generalization ability displayed by these foundation models presents opportunities for their application in Scan2BIM, which is the shorthand for Scan-to-BIM, for unknown scenarios. Nonetheless, the utilization of the vision foundation models' generalization capability and bridging the gap between the as-built data, including images and point clouds, and the as-built BIM, remain unexplored avenues of research.

To address the existing research gap, this study introduces Omni-Scan2BIM, a Scan2BIM approach for MEP scenes, utilizing vision foundation models. The proposed method enables the detection and modeling of essential MEP components without requiring additional training. The technique leverages two data sources: camera data and LiDAR point cloud data. The presented approach consists of four modules. In the first module, an all-purpose vision foundation model extracts features from a sample image containing the target component, as well

as other on-site images. These features are used to calculate similarities between the specified object and the images to be processed, generating similarity maps in the second module. Local maxima of the similarity map are sampled to obtain prior points, which serve as input prompts for the segmentation model in the third module. The promptable segmentation model segments the target objects from the on-site images in a sequential manner using prior points. In the fourth module, the transformation matrices between the camera coordinate system and the world coordinate system are employed to segment the target objects from the LiDAR point clouds, utilizing the 2D segmentation masks. The skeletons of MEP components are extracted, and parametric BIMs are reconstructed based on these extracted skeletons. Section 2 provides a comprehensive literature review of related works on as-built BIM generation using 2D images and 3D point clouds. Section 3 describes the developed approach for automatic creation of as-built BIMs for MEP scenes. A validation test on real construction scenarios and performance analysis are conducted in Section 4. Finally, Section 5 summarizes the study and offers suggestions for future work.

2. Literature review

2.1. Semantic-rich 3D point cloud generation

The acquisition of semantic-rich 3D point clouds plays a crucial role in the Scan2BIM process as it enables the creation of an accurate and information-rich BIM model. Precisely identifying object boundaries within the point cloud data is a fundamental and indispensable stage in this process. Two intuitive approaches can be employed to generate such semantic-rich 3D data. The first approach involves mapping the results obtained from 2D segmentation into 3D space, while the second approach directly conducts 3D segmentation on point clouds. Currently, deep learning methods have emerged as the predominant paradigm in both 2D and 3D semantic segmentation tasks.

Since the introduction of fully convolutional networks (FCN) in 2014 [14], significant advancements have been made in 2D image semantic segmentation. FCN models typically follow an encoder-decoder architecture, where the encoder is responsible for learning feature representations, and the decoder performs pixel-level classification based on these representations. Over the years, the encoder component has undergone evolution from AlexNet [15] and VGGNet [16] to ResNet [17], with ResNet becoming a widely adopted choice due to its effectiveness in feature representation. More recently, the Transformer architecture has emerged as a promising alternative to convolutional operators, exhibiting superior performance in various vision tasks. Transformers have the ability to capture long-range dependencies and increase the receptive field to encompass the entire image. This characteristic has led to the development of novel semantic segmentation models built upon transformer blocks, such as SETR [18], SegFormer [19] and Segmenter [20]. These models showcase improved performance and demonstrate the potential of transformer-based encoders in semantic segmentation tasks. To utilize 2D segmentation results for modeling purposes, mapping of these results into 3D space is necessary. This can be achieved using either the Structure from Motion (SfM) technique or the Simultaneous Localization and Mapping (SLAM) system.

In comparison to 2D methods, direct segmentation of 3D point clouds is a more straightforward approach. PointNet [21] pioneered the application of deep learning methods for processing point cloud data. It employed two symmetric functions, namely multilayer perceptron (MLP) and maxpooling, for feature extraction. However, PointNet did not consider the local dependency among points, making it less robust for detailed segmentation tasks. To address this limitation and explore local point features, more advanced models have been developed, such as PointNet++ [22], RandLA-Net [23] and KP-FCNN [24]. More recently, addressing the challenge of intensive 3D labeling, data-efficient learning for 3D point clouds has gained popularity. Representative works in this field include PointContrast [25], Contrastive Scene

Contexts [26] and SQN [27]. These methods achieve comparable performance to fully supervised methods while utilizing significantly fewer labeled samples for training. By leveraging contrastive learning and scene context information, these approaches enabled more efficient use of labeled data, reducing annotation efforts and increasing scalability.

Regardless of whether 2D or 3D semantic segmentation methods are employed, one prominent challenge is the limited generalization ability of pretrained models. It is notoriously difficult to directly apply a pretrained model to unseen scenarios. For instance, a well-trained model on indoor scenes may exhibit poor performance when applied to industrial or construction sites without retraining or fine-tuning on a customized dataset. Moreover, even when a model is trained on a specific industrial plant, it may not function as expected when deployed in another plant due to variations in appearances and differences in objects between the two scenes. This lack of generalization poses a significant obstacle for practical implementation of Scan2BIM solutions.

2.2. As-built BIM reconstruction for MEP systems

The MEP systems in the construction industry possess three significant characteristics: high complexity, frequent design changes, and prevalent installation deviations. Consequently, the AEC industry places great emphasis on establishing accurate as-built models of MEP systems. Currently, advanced algorithms have emerged with the potential to assist in the Scan2BIM process. Based on the authors' previous articles [28,29], MEP components can be broadly categorized into two groups: regular-shaped components and irregular-shaped components. Regular-shaped components, also known as tubular components, are characterized by constant cross sections and varying lengths. Examples include circular pipes and ducts. Irregular-shaped components, on the other hand, refer to components with complex shapes such as valves and pumps. The shapes of these components are fixed for a specific model type, and for the purpose of BIM reconstruction, six parameters $[x, y, z, p_x, p_y, p_z]$, which include locations and poses, are required.

For regular-shaped components, the majority of previous research has focused on the detection of cylindrical pipes. Two widely explored methods include the generalized Random Sample Consensus (RANSAC) and the centerline-based approach. Schnabel et al. [30] developed an efficient RANSAC-based method capable of detecting various simple 3D shapes, such as planes, spheres, cylinders, cones, and tori. This contribution has laid a solid foundation for Scan2BIM research endeavors. In the centerline-based method, tubular pipes are succinctly represented by their skeleton diagrams, which can be obtained through section analysis or Hough voting in a Gaussian sphere. Ahmed et al. [31] employed a two-dimensional Hough Transform on sliced point clouds to identify pipes with predetermined sizes. Bosché et al. [32] integrated the "Scan-to-BIM" and "Scan-vs-BIM" frameworks based on section analysis, enabling efficient construction monitoring of MEP components. Wang et al. [28] developed a slicing method wherein point cloud slices were analyzed, and the cross-sections of desired components could be extracted using two-dimensional approaches. In addition to slicing method, Gaussian sphere is frequently utilized to determine the principal directions of cylinders [33–35]. Patil et al. [36] further enhanced this methodology by introducing hierarchical accumulation structures in the Hough voting process. Furthermore, Lee et al. [37] extracted pipeline skeletons by employing Voronoi diagrams and topological thinning, subsequently classifying the skeletons into straight pipes, elbows, and tee pipes.

In recent years, there has been a growing trend in adapting generalized deep learning models for the purpose of pipeline extraction. Cheng et al. [38] developed DeepPipes, a customized deep learning model designed to reconstruct 3D pipelines from point clouds. Their approach utilized convolutional networks to learn point cloud features and classified points into different classes. Subsequently, robust clustering techniques and graph-based aggregation methods were applied to compute a coherent pipe model. Xie et al. [39] proposed a neural

network called PipeNet, which incorporated an optimal neighborhood selection module for pipe segmentation and parameter prediction. This network aimed to accurately segment and predict parameters for pipes. Qi et al. [40] introduced the DSCNet, a deep learning model equipped with a novel dynamic snake convolution. This convolutional operation enabled the network to effectively capture the features of tubular structures by adaptively focusing on slender and tortuous local structures. The integration of this dynamic snake convolution significantly enhanced the accuracy and continuity of tubular structure segmentation. However, it is still challenging of achieving Scan2BIM for ducts with non-circular cross sections and pipes with small radii. Very few existing methods have effectively addressed this issue.

For irregular-shaped components with complex geometries, such as valves and pumps, feature-based methods are often preferred. Deep learning-based descriptors, in comparison to hand-crafted point cloud descriptors, are more expressive and provide comprehensive information, exhibiting significantly improved performance in various tasks, including semantic segmentation and object recognition. Agapaki and Brilakis [41] developed a framework named "CLOI-NET" based on PointNet++ to automatically annotate industrial point clouds. Likewise, [42,43] proposed deep learning frameworks for point cloud semantic segmentation specifically tailored to MEP scenes. Wang et al. [29] initially detected target objects in 2D images and subsequently refined the detection results in 3D space. To enhance the reliability of 3D verification methods based on the Iterative Closest Point (ICP) algorithm and optimization-based techniques, Wang et al. [44] considered local shape similarity and developed an object verification module. This module can be easily integrated into any Scan2BIM framework, leading to improved accuracy in the Scan2BIM process.

However, similar to generalized deep learning-based object detection algorithms, the limited generalizability of these methods hinders their application in real-world projects. Furthermore, 3D deep learning methods encounter challenges in capturing fine details of MEP piping systems and recognizing subsystems.

2.3. Vision foundation model

Pre-trained on extensive web-scale datasets, LLMs such as ChatGPT [13] have brought about a significant revolution in the field of natural language processing (NLP). These foundational models exhibit remarkable transfer capabilities, extending their effectiveness to tasks and data distributions that lie beyond their original training scope. LLMs demonstrate impressive zero-shot and few-shot generalization abilities and excel in addressing diverse language-related tasks, including language understanding, generation, interaction, and reasoning.

The research on foundation models in computer vision is rapidly advancing and catching up with the progress made in NLP. Motivated by large-scale image-text contrastive pre-training, models such as CLIP [45] and ALIGN [46] demonstrate robust zero-shot transfer capabilities to downstream vision tasks. In contrast to being guided by text, DINOv2 [47] focuses on learning versatile visual features by capturing intricate information at both the image and pixel level solely from raw image data. It achieves comparable or superior performance compared to CLIP on downstream tasks. However, the utilization of these foundation models as image encoders necessitates task-specific heads for individual downstream tasks, which limits their generalizability in real-world applications. Recently, Segment Anything Model (SAM) [48] has exhibited impressive zero-shot segmentation performance, showcasing significant potential in open-world image perception. SAM introduces a novel promptable segmentation paradigm, where a handcrafted prompt is provided as input, and the model returns the corresponding mask. SAM supports a wide range of acceptable prompts, including points, boxes, masks, and free-form texts, enabling the segmentation of various visual elements. Nevertheless, as a class-agnostic segmenter, SAM lacks the ability to extract high-level semantic features, which restricts its capabilities for comprehensive open-world image understanding.

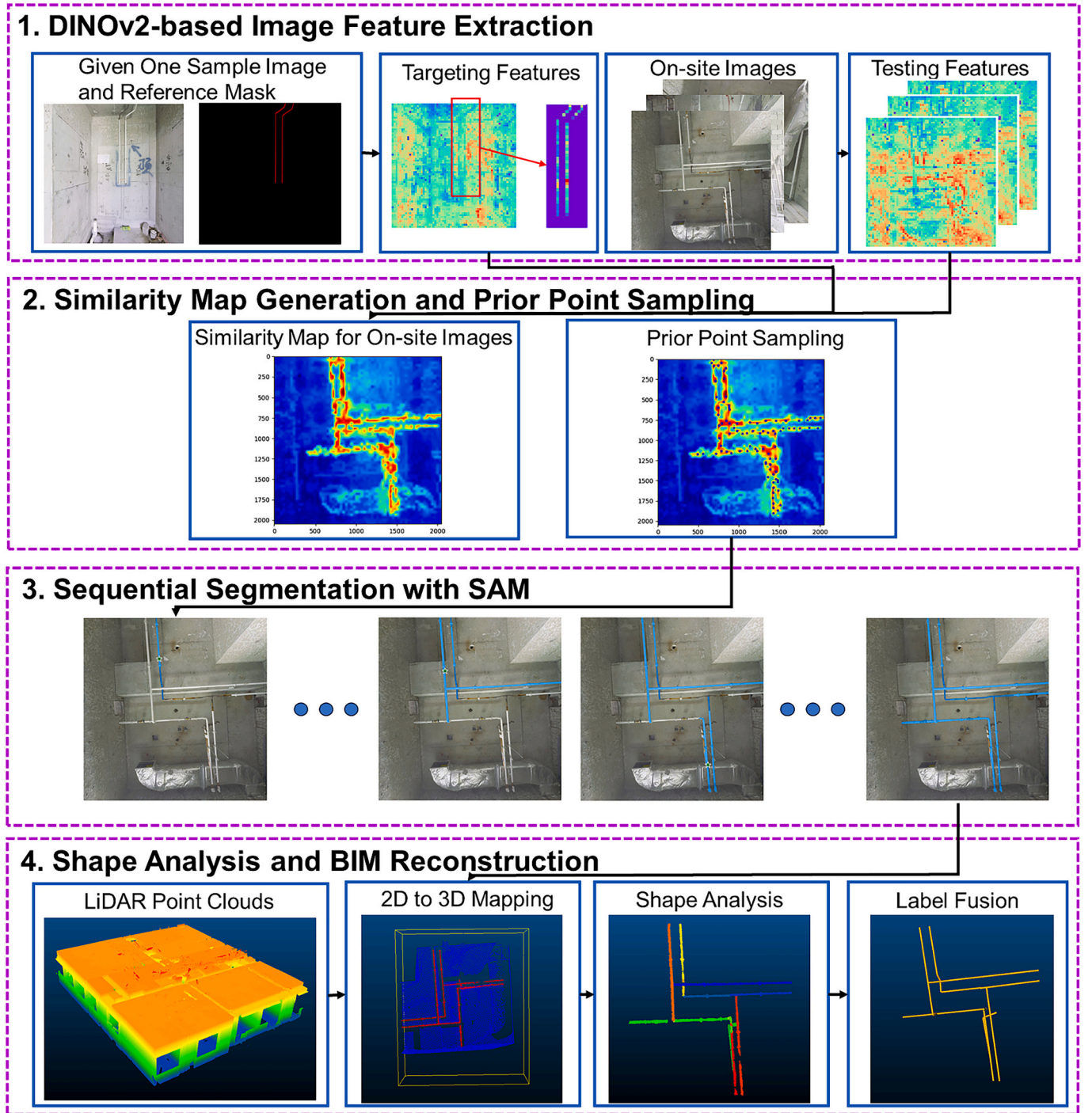


Fig. 1. Overview of the proposed Omni-Scan2BIM based on vision foundation models.

In recent years, researchers have been exploring the integration of various foundation models to address the aforementioned limitations. Grounded-SAM [49] is an example of such efforts, which combines Grounding DINO with Segment Anything to establish a robust pipeline for open-world image understanding. Building upon SAM, Zhang and Liu [50] proposed a general solution called SAMed for medical image segmentation. They employed a low-rank-based (LoRA) fine-tuning strategy on the SAM image encoder and jointly finetuned it with the prompt encoder and the mask decoder using labeled medical image segmentation datasets. To eliminate the need for manual prompting steps, Zhang et al. [51] proposed a training-free approach for personalizing SAM, referred to as PerSAM. PerSAM incorporated three

modules: target-guided attention, target-semantic prompting, and cascaded post-refinement. Furthermore, Liu et al. [52] introduced a bidirectional matching strategy for accurate cross-image semantic dense matching and a robust prompt sampler for mask proposal generation.

However, the direct application of previous promotable SAM approaches to MEP networks results in unsatisfactory segmentation outcomes due to the presence of thin local structures and variable global morphologies within the MEP components. Additionally, the utilization of vision foundation models to bridge the gap between as-built data, comprising images and point clouds, and as-built BIMs remains largely unexplored. Consequently, there is a need to investigate how to harness the generalization abilities of vision foundation models and address the

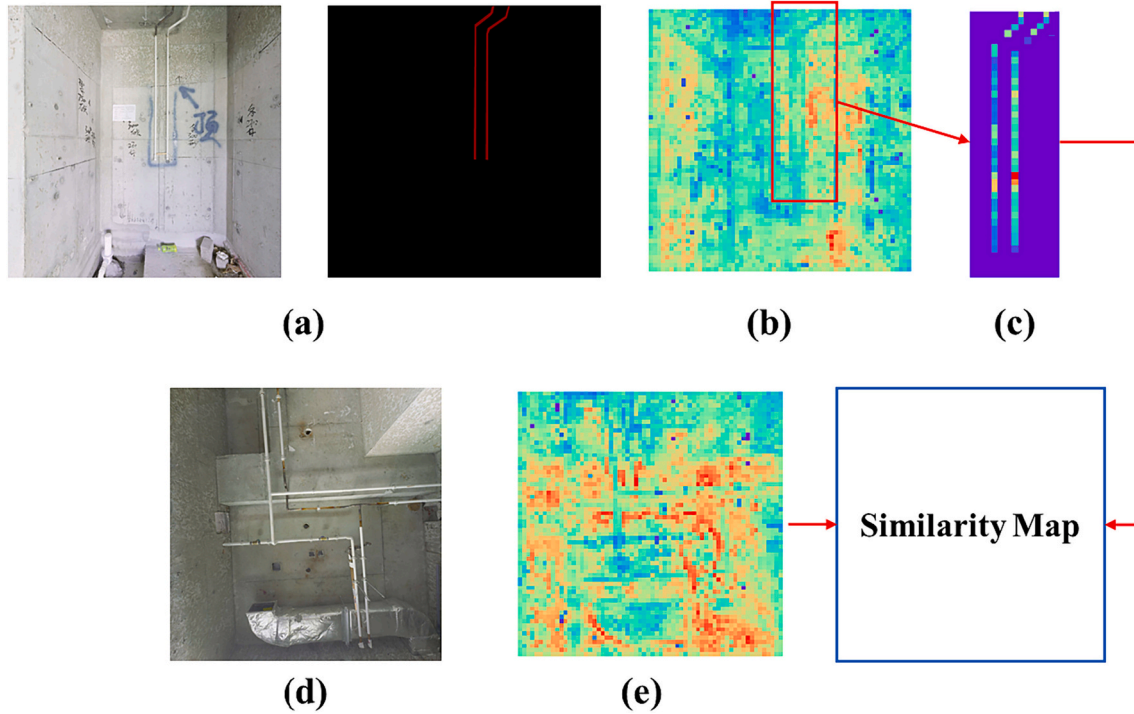


Fig. 2. Foundation model based feature extraction. (a) The sample image containing the target object along with its corresponding mask; (b) The pixel-level features extracted for the sample image; (c) The pixel-level features specifically related to the target object; (d) An on-site image to be processed; (e) The pixel-level features extracted for the on-site image.

challenges associated with transforming as-built data into accurate as-built BIMs. To unlock the full potential of vision foundation models in the Scan2BIM process, this study proposes an approach called Omni-Scan2BIM. The proposed method aims to convert as-built data, including on-site images and point clouds, into as-built BIMs without requiring explicit training. Remarkably, the Omni-Scan2BIM approach achieves this goal by leveraging a single sample image containing the target components, along with a reference mask, as the input. This innovative approach demonstrates the potential to streamline the Scan2BIM process and eliminate the need for extensive training datasets.

3. Methodology

The objective of this study is to present Omni-Scan2BIM, a ready-to-use Scan2BIM approach that does not require training or fine-tuning on a custom dataset. By utilizing a single sample image containing the target components along with a reference mask, the proposed method automatically recognizes all target components within the entire site. Fig. 1 provides an overview of the Omni-Scan2BIM approach. The method comprises four sequential modules: 1) foundation model-based feature extraction, 2) similarity map generation and prior point sampling, 3) sequential class-agnostic segmentation, and 4) shape analysis and BIM reconstruction. These modules will be explained in Sections 3.1 to 3.4, respectively. In Section 3.1, pixel-level deep descriptors are computed for the target object present in the sample image as well as other on-site images. Then, in Section 3.2, a comparison of visual features between the target object and the on-site images is performed, resulting in the generation of similarity maps for the on-site images. These similarity maps indicate the probability of each pixel belonging to the target category. Subsequently, prior points are sampled from the local maxima of the similarity map. In Section 3.3, the prompt encoder of SAM is employed to sequentially process the prior points and perform segmentation of the contiguous region surrounding each prompt point. Finally, in Section 3.4, leveraging the transformation matrix between the camera coordinate and the world coordinate, the 2D segmentation

masks are projected to 3D space, and the skeletons of the tubular objects are extracted. Based on geometric analysis, the as-built BIMs are reconstructed in a parametric manner.

3.1. Foundation model based feature extraction

Due to limitations in model capacity and the diversity of datasets, traditional deep convolutional neural networks (DCNNs) can only generate reliable feature vectors for objects present in the training dataset. Consequently, the application of deep learning in unknown environments is hindered. However, recent advancements in the field, such as the Transformer architecture and the availability of expansive databases, have enabled the development of vision foundation models capable of encoding comprehensive features for previously unseen objects through large-scale pre-training. One notable example is the Masked and Asymmetrically Encoded (MAE) model [53], which employs an asymmetric encoder-decoder structure and conducts masked image modeling to efficiently and effectively train scalable vision Transformer models. MAE has demonstrated exceptional fine-tuning performance across various downstream tasks. Another noteworthy model is Contrastive Language-Image Pre-training (CLIP) [45], which learns image representations from scratch using a dataset of 400 million image-text pairs. CLIP exhibits impressive zero-shot image classification capabilities. Additionally, the DINOv2 [47] model utilizes image and patch level discriminative self-supervised learning to acquire all-purpose visual features suitable for diverse downstream tasks. Notably, DINOv2 excels in patch-matching, capturing information related to semantic parts that share similar characteristics across different objects. Based on the technical reports [47,52], DINOv2 is the most suitable image encoder capable of providing highly reliable features among different vision models. Consequently, in the Omni-Scan2BIM framework, DINOv2 has been utilized to extract features from both the reference object and the on-site images, ensuring the robustness and accuracy of the feature extraction process.

For convenience, the sample image and the corresponding mask are

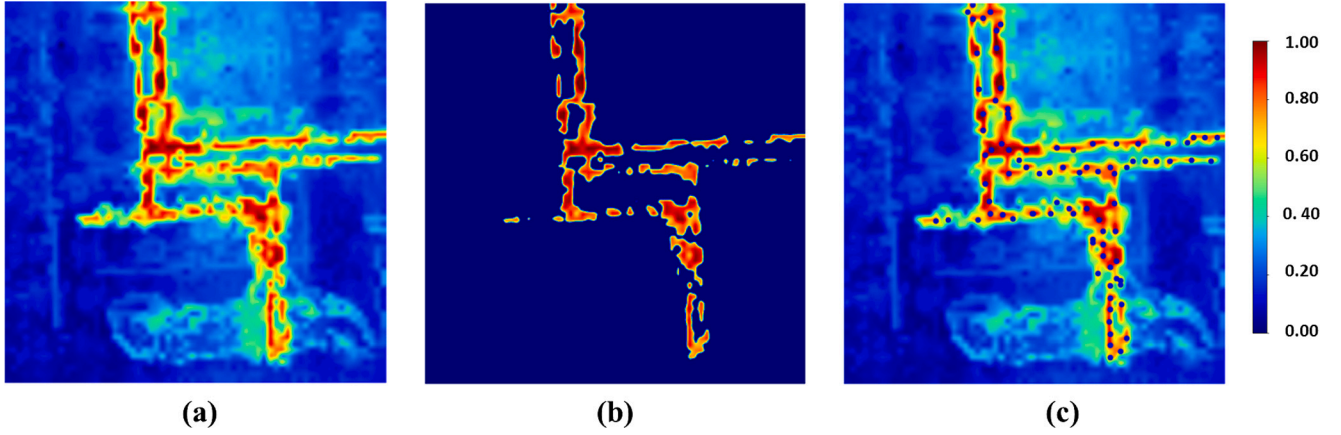


Fig. 3. Similarity map generation and prior point sampling. (a) Similarity map generated through feature comparison for the on-site image; (b) Similarity map after eliminating pixels with low similarity values. (c) Prior point sampling based on the similarity map.

denoted as I_s and M_s respectively, as depicted in Fig. 2(a). The on-site images to be processed are represented as I_t , as illustrated in Fig. 2(d). The encoding operation performed by the foundation vision model is denoted as ENC_{DINOv2} . The visual features of the sample image, denoted as F_s and shown in Fig. 2(b), and the on-site images, denoted as F_t and illustrated in Fig. 2(e), are calculated using the following formulas:

$$F_s = ENC_{DINOv2}(I_s), F_t = ENC_{DINOv2}(I_t) \quad (1)$$

where $F_s, F_t \in \mathbb{R}^{h \times w \times c}$. h and w denote the spatial dimensions, and c represents the number of channels. To eliminate the influence of the background, the visual features of the target object, F_o , are obtained by spatial-wise multiplying the sample image features, F_s , with the reference mask, M_s :

$$F_o = M_s \circ F_s \quad (2)$$

where $\circ$ denotes spatial-wise multiplication. The resulting feature representation, $F_o \in \mathbb{R}^{d \times c}$ is visualized in Fig. 2(c), where d corresponds to the size of the object in the feature map F_s .

3.2. Similarity map generation and prior point sampling

In Omni-Scan2BIM, an efficient method is adopted to generate the similarity map for the on-site images: Firstly, the features of the target object, F_o , are aggregated using an average pooling operation:

$$E_o = Pooling(F_o) \quad (3)$$

where $E_o \in \mathbb{R}^{1 \times c}$ represents the compressed global visual embedding of the target object. Secondly, the features of the on-site images, F_t and the global embedding of the target object, E_o , are L2-normalized. Thirdly, the cosine similarity matrix, $S \in \mathbb{R}^{h \times w}$, is computed through matrix multiplication:

$$S = F_t E_o \quad (4)$$

After resizing and bilinear interpolation, the resulting similarity map is visualized in Fig. 3(a). In the similarity maps, the jet colormap was employed, where red indicates high similarity and blue represents low similarity. Ideally, higher similarity values indicate a greater likelihood of each pixel belonging to the target category. In practice, this can be challenging due to variations in appearance or even differences in object categories between the reference and target objects. Therefore, it is crucial for the encoder to embed rich and detailed information within these features.

Upon acquiring the similarity map, a limited set of prior points needs to be generated to facilitate precise target object segmentation using the promptable segment anything model SAM. Initially, pixels exhibiting a

similarity value below a specified threshold are identified and subsequently eliminated as background. The resulting similarity map, obtained after this initial filtering process, is illustrated in Fig. 3(b). Considering the potential variation of the threshold value across different applications, it is advisable, particularly when recognizing a new type of targeting object in novel scenes, to fine-tune this value within a localized region to ensure its suitability for the specific application. Once determined, this refined threshold value can be consistently applied throughout the entire project. Based on the authors' tests, a threshold value of 0.5 is suggested as a favorable starting point to initiate this fine-tuning process. Following the filtering step, regional maxima are detected within the processed similarity map presented in Fig. 3(b) to serve as prior points. These maxima correspond to pixels where the similarity values are relatively higher, surrounded by pixels displaying lower similarity values. For enhanced clarity, the sampled prior points are superimposed on the original similarity map, as illustrated in Fig. 3(c). The search for regional maxima could be carried out utilizing functions from the Scikit-Image [54], an open-source image processing library. The sampled prior points are depicted in Fig. 3(c).

3.3. Sequential segmentation with SAM

Recently, SAM has gained attention for its impressive zero-shot class-agnostic segmentation performance, achieved through pre-training on a vast dataset of 1 billion masks and 11 million images using the model-in-the-loop annotation technique. SAM introduces a novel promptable segmentation task, where the objective is to generate a segmentation mask for any given prompt, including prior points or bounding boxes. However, SAM inherently lacks the ability to segment specific visual concepts. This necessitates users to manually locate the target object in various poses or contexts for each image and then activate SAM with a precise prompt for segmentation. This process becomes labor-intensive and time-consuming, particularly for object segmentation tasks in large-scale scenes. To address this limitation, Omni-Scan2BIM leverages the prior points generated through similarity analysis, as described in Sections 3.1 and 3.2, to enable SAM to automatically segment unique visual concepts in a simple and efficient manner.

The prior points are sorted in descending order based on their similarity values. The point with the highest similarity is initially fed to SAM for segmentation, as indicated by the green star in Fig. 4(a). SAM comprises three main components: an image encoder $ENC_{SAM-img}$, a prompt encoder $ENC_{SAM-prompt}$, and a lightweight mask decoder $DEC_{SAM-mask}$. In Omni-Scan2BIM, SAM takes an input image I_t and the prior point P with the highest similarity. Firstly, $ENC_{SAM-img}$ is employed to extract the image feature, while $ENC_{SAM-prompt}$ encodes the given prior point into c -dimensional tokens. Subsequently, the encoded image and

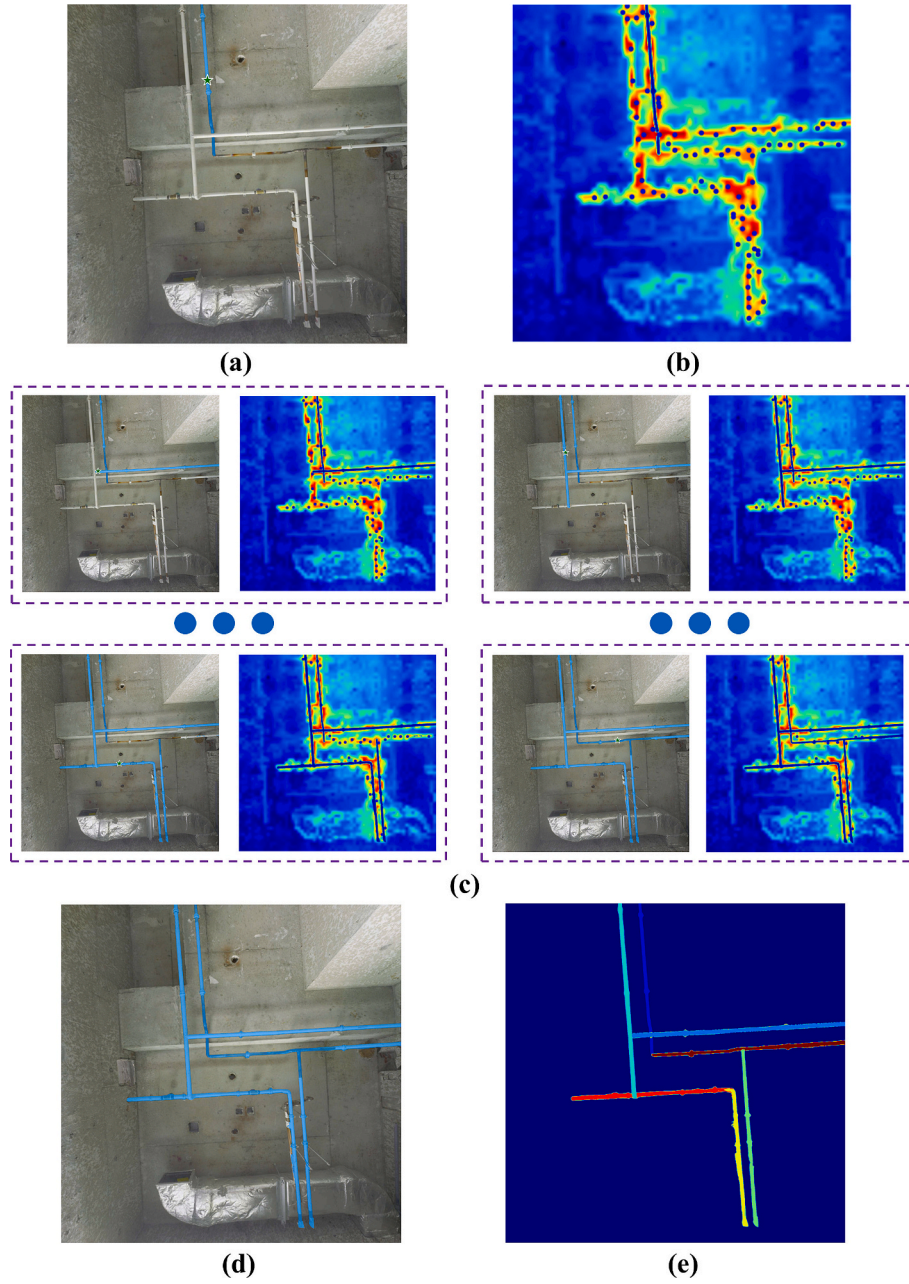


Fig. 4. Sequential segmentation using SAM. (a) Predicted mask after the initial segmentation with SAM, utilizing the prior point with the highest similarity value; (b) Remaining prior points after the first segmentation with SAM; (c) Sequential segmentation with the visualization of mask predictions and prior points; (d) The final mask for slim pipes; (e) The final mask for slim pipes with indices recorded.

the prompt tokens are fed into the decoder $DEC_{SAM-mask}$ for attention-based feature interaction. The input tokens of the decoder are constructed by concatenating several learnable tokens T_M as a prefix to the prompt tokens. These mask tokens play a crucial role in generating the final mask output. The segmentation process using SAM can be formulated as follows:

$$M = DEC_{SAM-mask} \left(ENC_{SAM-img} (I_I), Concat(ENC_{SAM-prompt} (P), T_M) \right) \quad (5)$$

where M represents the zero-shot mask prediction generated by SAM, as depicted in Fig. 4(a). In the original SAM framework, multiple masks of three scales are simultaneously generated, representing the entire object, its parts, and subparts. However, to mitigate ambiguity, Omni-Scan2BIM selects the mask with the highest confidence score, which is

estimated through Intersection over Union (IoU) calculation [48].

The segmentation result is initially evaluated using two straightforward metrics: the confidence score output by SAM and the segmentation purity. Segmentation purity is defined as the average similarity within the proposed mask region generated by SAM:

$$purity = \frac{\sum M \circ S}{\sum M} \quad (6)$$

where M represents the segmented Boolean mask, and S represents the similarity map.

Once the mask is obtained from SAM, it undergoes an appearance check step. If the confidence score from SAM and the segmentation purity exceeds the proper thresholds, the mask is deemed successful and passes the appearance check. It is then preserved for the subsequent shape-based filtering process, as elucidated in Section 3.4. Masks that

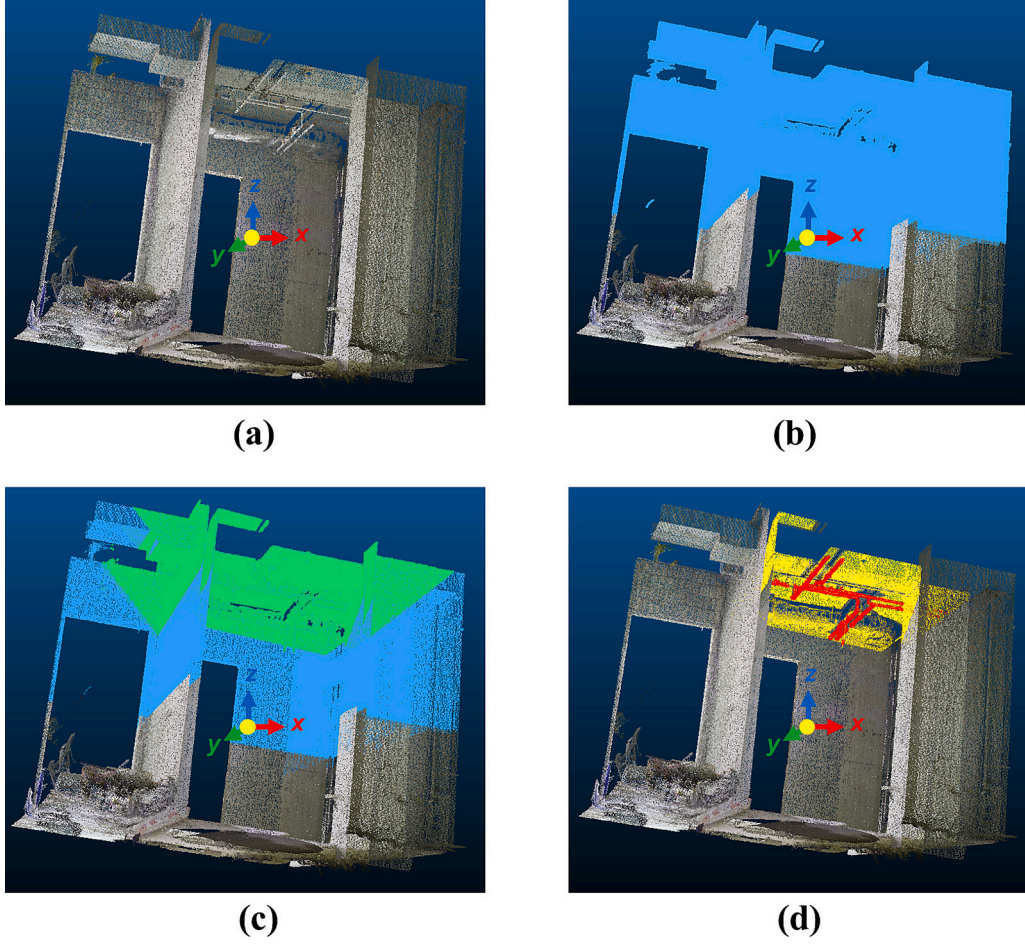


Fig. 5. 2D to 3D mapping. (a) The point cloud data in the camera coordinate system; (b) The blue points represent the points located on the frontside of the camera; (c) The green points represent the points situated within the camera's field of view (FOV); (d) The yellow points represent visible points of the camera and the red points specifically denote the target object. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

satisfy both appearance and shape requirements are retained for modeling purposes. Simultaneously, the prior points falling within the mask area are removed as shown in Fig. 4(b). Conversely, masks that fail to meet the aforementioned checkpoints are disregarded, and the corresponding prior point is eliminated. Following the initial segmentation and inspection, the prior point with the highest similarity in the remaining points is fed into SAM for a second-round of segmentation and filtering. This sequential process of segmentation and inspection continues iteratively until there are no more prompts available. The advantage of sequential segmentation of the target object, as opposed to segmenting the target object in a single iteration with multiple prior points, lies in its ability to effectively mitigate the level of noise contamination. The progression of this iterative segmentation process is visually represented in Fig. 4(c). The final mask specifically representing the slim pipes is shown in Fig. 4(d). Additionally, Fig. 4(e) provides a visual representation of the final mask for the slim pipes, with an accompanying index for reference.

3.4. Shape analysis and BIM reconstruction

3.4.1. 2D to 3D mapping

In this section, the qualified masks generated by SAM, which have successfully passed the appearance check, are mapped to 3D space for further analysis. This mapping is carried out for two primary reasons. First, since BIMs are essential 3D representations, it is necessary to map the semantic labels obtained from image segmentation to 3D space to facilitate BIM reconstruction process. Second, due to the inherent

complexity of real-world scenarios, masks that pass the initial appearance check may not necessarily represent the desired objects.

To transfer the labels of the 2D masks to the 3D point clouds, it is crucial to establish a clear transformation between the camera frame coordinate system and the world coordinate system. This transformation matrix can be obtained through meticulous hardware calibration. In the case of Omni-Scan2BIM, the built-in high dynamic range (HDR) cameras of commercial laser scanners are utilized to better illustrate the concept. The calibration of these cameras is performed by the manufacturers of the scanners. For each image captured by the built-in camera, the extrinsic parameters describing the transformation are recorded,

including a 3×3 rotation matrix $R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$ and a 3×1

translation vector $t = \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix}$.

The point clouds obtained from the laser scanner, represented ho-

mogeneously as $P_w = \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$ in the world coordinate system, are

initially transformed to the camera coordinate system denoted as $P_{cam} =$

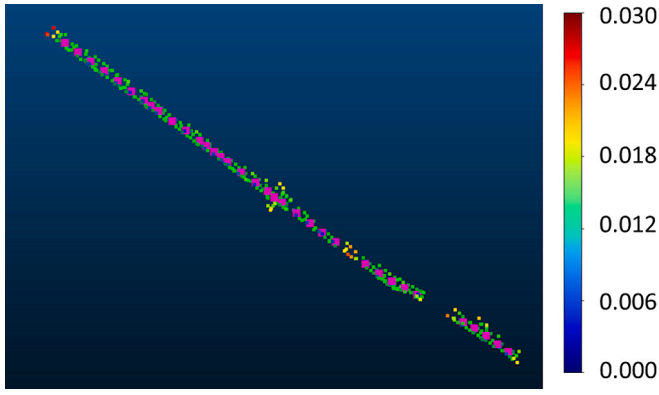


Fig. 6. Rule-based shape filtering for slim pipes. The extracted skeleton representing the slim pipe is depicted in a magenta colour, while the distance map is visualized using a jet colormap. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

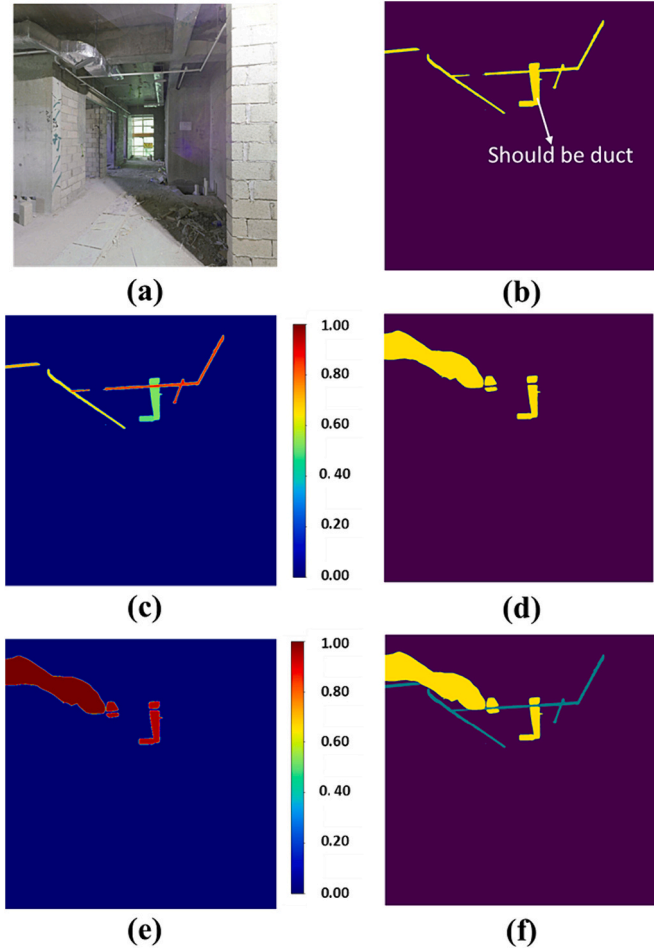


Fig. 7. Label fusion for integrating multiple classes within a single setup. (a) Raw on-site image; (b) Predicted mask for the slim pipe; (c) Certainty levels corresponding to the predicted mask for the slim pipe; (d) Predicted mask for the duct; (e) Certainty levels corresponding to the predicted mask for the duct; (f) Final mask representing the slim pipes and the ducts.

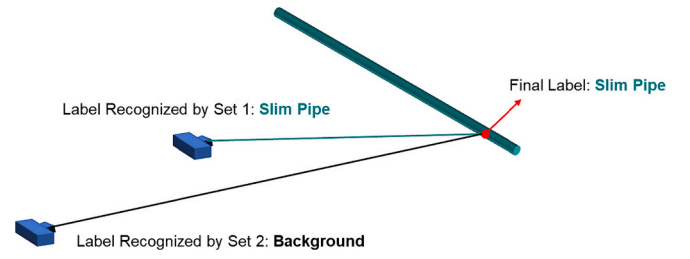


Fig. 8. Label fusion for integrating multiple setups.

using the transformation matrix:

$$\begin{bmatrix} X_{cam} \\ Y_{cam} \\ Z_{cam} \\ 1 \end{bmatrix}$$

$$P_{cam} = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} P_w = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad (7)$$

The transformed point cloud, P_{cam} , is illustrated in Fig. 5(a). Subsequently, points located behind the camera, specifically those with a Z value less than 0 are removed. The remaining points, depicted in Fig. 5 (b), are visually distinguished by their blue colour.

Following this, the transformed point cloud, P_{cam} , is mapped to the image plane denoted as $P_{img} = \begin{bmatrix} X_{img} \\ Y_{img} \\ Z_{img} \end{bmatrix}$ using the projection matrix $K =$

$\begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$, which incorporates the camera's intrinsic parameters such as focal length in the x-axis (f_x), focal length in the y-axis (f_y), offset in the x-axis (o_x) and offset in the y-axis (o_y):

After normalizing the homogeneous coordinates, the pixel coordinates in the image plane, $P_{pxl} = \begin{bmatrix} u \\ v \end{bmatrix}$, are calculated as follows:

$$P_{img} = KP_{cam} = \begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X_{cam} \\ Y_{cam} \\ Z_{cam} \\ 1 \end{bmatrix} \quad (8)$$

Upon coordinate transformation, the points within the camera's field of view (FOV) are expected to fall within the boundaries of the captured image, adhering to geometric principles:

$$u = \frac{X_{img}}{Z_{img}}, v = \frac{Y_{img}}{Z_{img}} \quad (9)$$

where W and H represent the width and height of the captured image, respectively. Points that do not meet these specified criteria are eliminated. The retained points are highlighted in green in Fig. 5(c).

$$0 < u < W, 0 < v < H \quad (10)$$

Up to this point, it has been observed that certain points are erroneously included in the image, despite being occluded and not visible. For example, points located behind interior walls may also be erroneously included. To address this issue and remove hidden points, a visibility analysis is conducted by extracting the points residing on the convex hull of the transformed point cloud, as presented in [55]. The points highlighted in yellow in Fig. 5(d) indicate the visible points within the camera's view.

Following the projection, each visible point is mapped to a pixel in

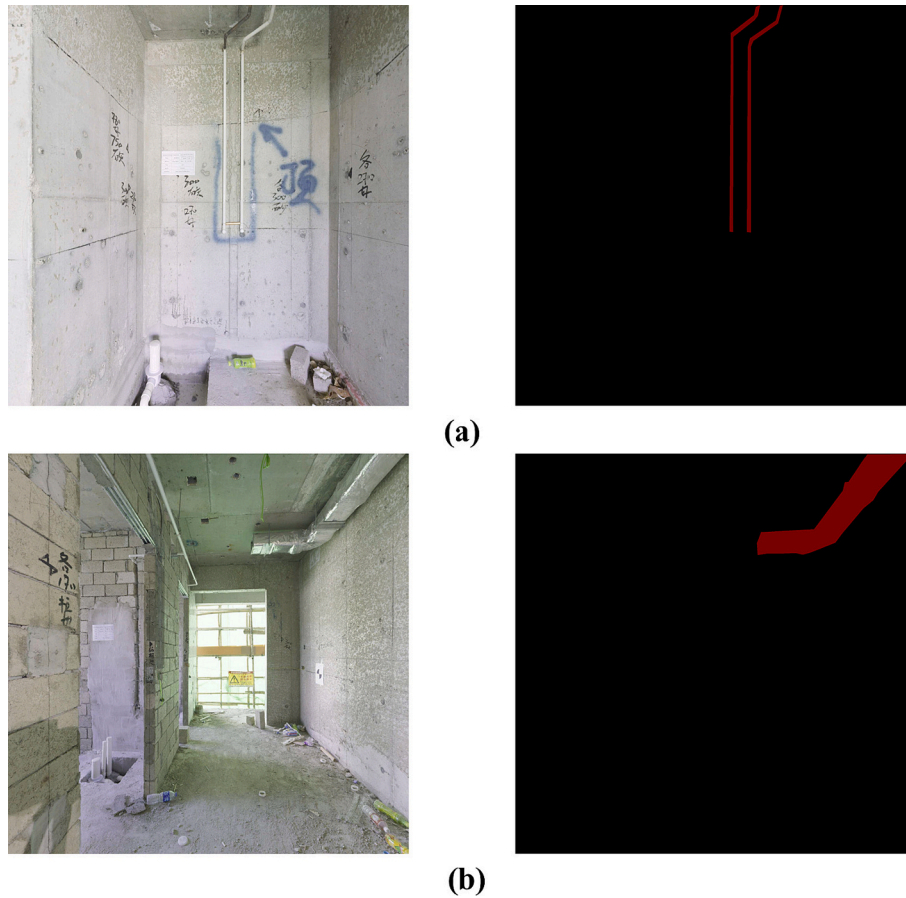


Fig. 9. Sample images for slim pipes and ducts. (a) The sample image and the manually labeled mask for slim pipes; (b) The sample image and the manually labeled mask for ducts.

the mask proposed by SAM in Section 3.3, and the corresponding label is assigned to that point. This process effectively segments the target objects in 3D space, as exemplified by the distinct red coloration in Fig. 5 (d). At this stage, the connection between the image and the 3D point clouds has been established, signifying that accurate segmentation in 2D space is equivalent to accurate segmentation in 3D space. Consequently, the 3D objects can be successfully segmented for further analysis.

3.4.2. Rule-based shape filtering

Given that there are still a specific number of undesired masks that have passed the appearance check, which includes SAM segmentation confidence check and purity check, it becomes imperative to devise a filtering strategy as the authors' previous works [28,29] to enhance the quality of the segmentation results. As this study focuses on regular shaped components with intricate global morphology, the discussion regarding the shape filtering strategy for irregular shaped components is beyond the scope of this research. Interested readers are encouraged to refer to the authors' previous work [44] for further insights into this topic.

Regular shaped components are characterized by continuous cross sections and varying lengths, with circular pipes, rectangular ducts, and I-beams being the most representative examples. The design of filtering rules should be tailored to the distinctive features exhibited by these components. Notably, the segmentation output achieved with the aid of the powerful SAM algorithm in the newly proposed Omni-Scan2BIM system demonstrates significantly improved cleanliness compared to the rough segmentation steps presented in the authors' earlier works [28,29]. Consequently, the filtering step employed in this study showcases enhanced intuitiveness and efficiency.

1) Circular Pipes

The distinctive characteristic of circular pipes lies in their cylindrical bodies, allowing for the seamless integration of the efficient RANSAC algorithm [30] to fit the segmented point clouds. Subsequently, the average distance between the point clouds and the fitted cylinders can be computed, enabling the identification and removal of false positives that surpass an appropriate threshold. The current methodology offers two notable advantages over the authors' previous approaches. Firstly, the Omni-Scan2BIM system demonstrates enhanced efficiency. In the authors' earlier works, alongside circular pipes, there were substantial amounts of extraneous point clouds representing irrelevant objects following the rough segmentation operation. To ensure the robustness of the entire framework, additional cross-section analysis was performed consecutively in the X, Y, and Z directions within the Manhattan coordinate system. However, this process proved to be time-consuming. Secondly, the versatility of the Omni-Scan2BIM system allows for the accurate representation of tubular, elongated objects with arbitrary major directions, achieved through a more comprehensive 2D analysis and 3D filtering approach.

For extremely thin pipes with a diameter less than three centimeters, the effectiveness of the RANSAC algorithm diminishes due to the sparsity of points, resulting in blurry cross-sectional representations that pose significant challenges in identifying circular shapes. To address this complex issue, the local width of the object is estimated to filter out wrong recognitions based on the assumption that slim pipes are the only object with a local width smaller than three centimeters. In line with this idea, the 3D skeleton is extracted using the Laplacian-based contraction (LBC) method [56] for the candidate 3D point clouds, as depicted in Fig. 6. Similar to circular pipes, a distance field can be calculated from

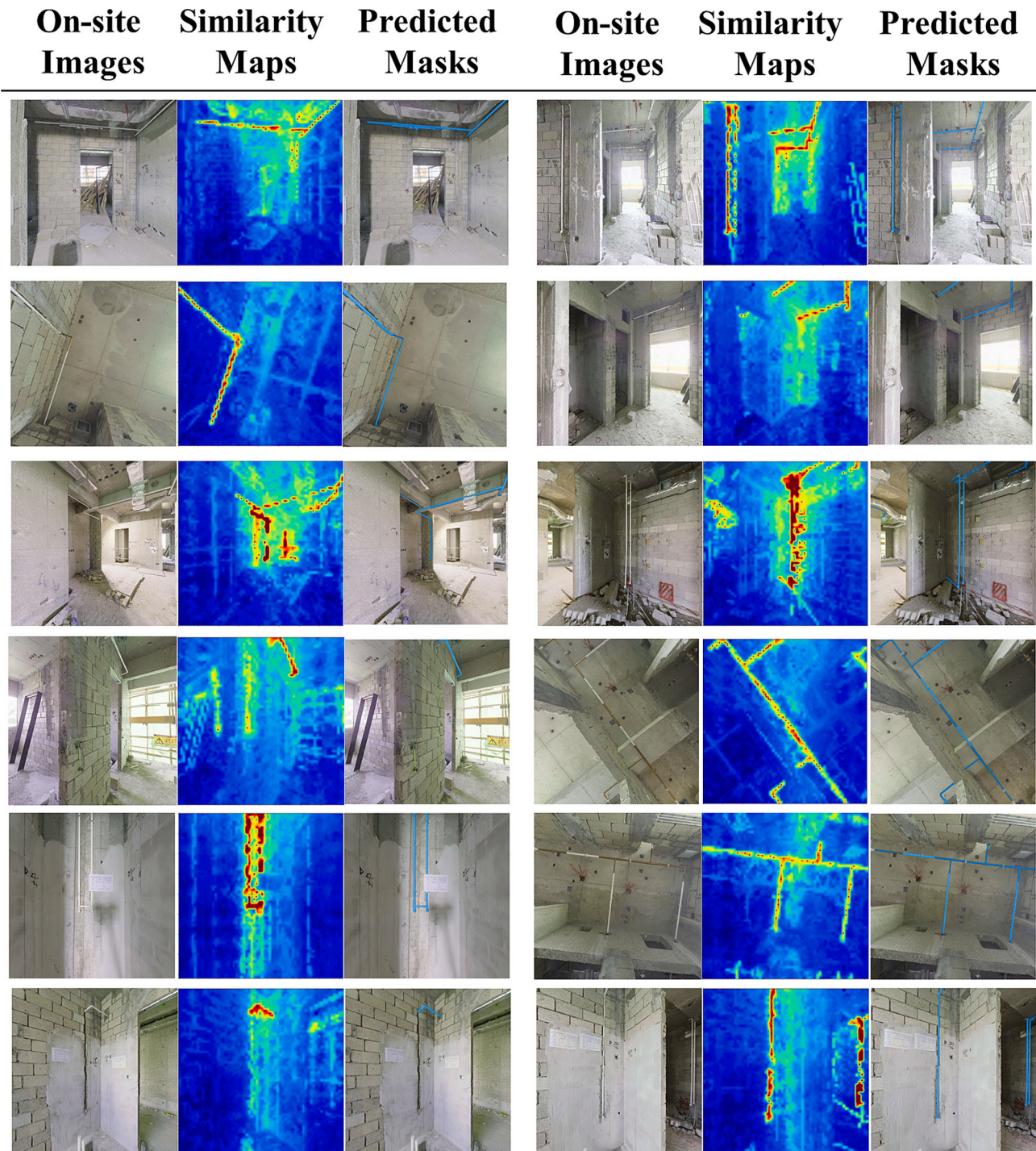


Fig. 10. Visualization of similarity maps and predicted masks for slim pipes.

the point clouds to the extracted skeleton, as illustrated in Fig. 6. Any point sets with distances that exceed an appropriate threshold are considered false positives and are subsequently discarded.

2) Rectangular Ducts

In contrast to circular pipes, the verification of rectangular ducts cannot be effectively accomplished using the efficient RANSAC method. Instead, a cross section verification method can be employed. Firstly, the ducts' skeletons are extracted using the LBC technique. Secondly, the major direction of the extracted skeleton is determined using RANSAC algorithms. Thirdly, five cross sections along the major direction are

retrieved, and the width of each cross section is verified accordingly. It is important to note that the cross section verification process differs from the cross section analysis employed in the authors' previous works [28,29]. The main distinctions lie in the fixed number of cross sections utilized for verification in Omni-Scan2BIM, as opposed to the dense slicing approach employed in previous works, which involved a larger number of cross sections.

3.4.3. Label fusion and BIM reconstruction

Each on-site image may contain multiple types of components that need to be recognized. Since different types of components are processed independently, one potential error in recognition is misclassifying a

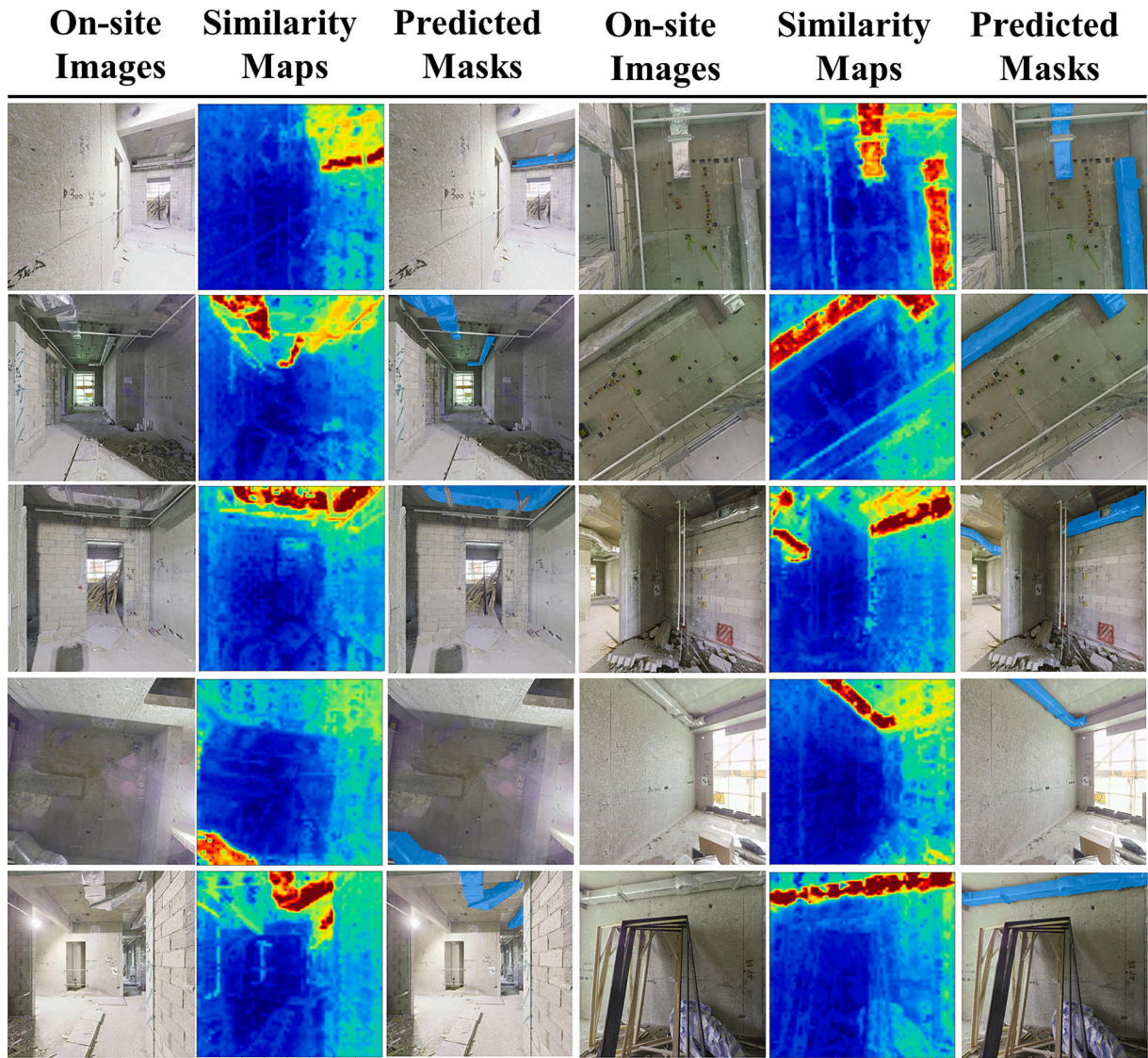


Fig. 11. Visualization of similarity maps and predicted masks for ducts.

component as a different class. An example of such an error is depicted in Fig. 7. Fig. 7(a) shows an on-site image where both slim pipes and ducts are expected to be recognized. Utilizing the techniques described in previous sections, the predicted masks for slim pipes and ducts are visualized in Fig. 7(b) and Fig. 7(d) respectively. It can be observed in Fig. 7(b) that one segment of the duct has been incorrectly recognized as a slim pipe.

To address this issue, a label fusion mechanism has been developed to handle multiple classes within a single setup. During the sequential segmentation process, for each segment that passes both the appearance check and the shape check, the similarity value of the prior point for that particular segmentation is recorded along with the segmented masks, indicating the certainty level of the proposed mask. The certainty levels for slim pipes and ducts in the captured on-site image are visualized in Fig. 7(c) and Fig. 7(e) respectively. It is evident that although the duct has been mislabeled as a slim pipe in Fig. 7(b), the certainty level for that segment is relatively low, as indicated by the light green colour in Fig. 7(c). Conversely, the certainty level for it being a duct is significantly higher, as shown in Fig. 7(e). In such cases, the pixels belonging to that segment will be labeled as ducts, as the certainty level of it being a duct is greater than that of it being a slim pipe. The resulting 2D mask after label fusion within one setup is displayed in Fig. 7(f).

At the current stage, the introduced approach can only be applied to

one image frame. However, considering the extensive area of the entire site to be modeled, which requires dozens to hundreds of stations to cover, the integration of segmentation masks from multiple setups through label fusion becomes imperative to achieve a unified result.

In the registered point cloud data, a single point may be visible from multiple laser scanning stations, which can result in conflicting labels assigned to that point. To address potential label conflicts, the Omni-Scan2BIM methodology employs the principle of proximity, whereby the label assigned to a point is determined by the prediction of the nearest scanning station as shown in Fig. 8. The data fusion strategy relies on the observation that objects closer to the camera exhibit greater clarity and are easier to recognize.

The BIM reconstruction process in this study adheres to the parametric modeling approaches previously introduced by the authors in their prior works [28,29].

4. Validation

4.1. Dataset for validation experiment

To assess the effectiveness of the proposed Omni-Scan2BIM method, point cloud data were collected using the LEICA RTC360 laser scanner at construction site of a university student apartment in Hong Kong. The

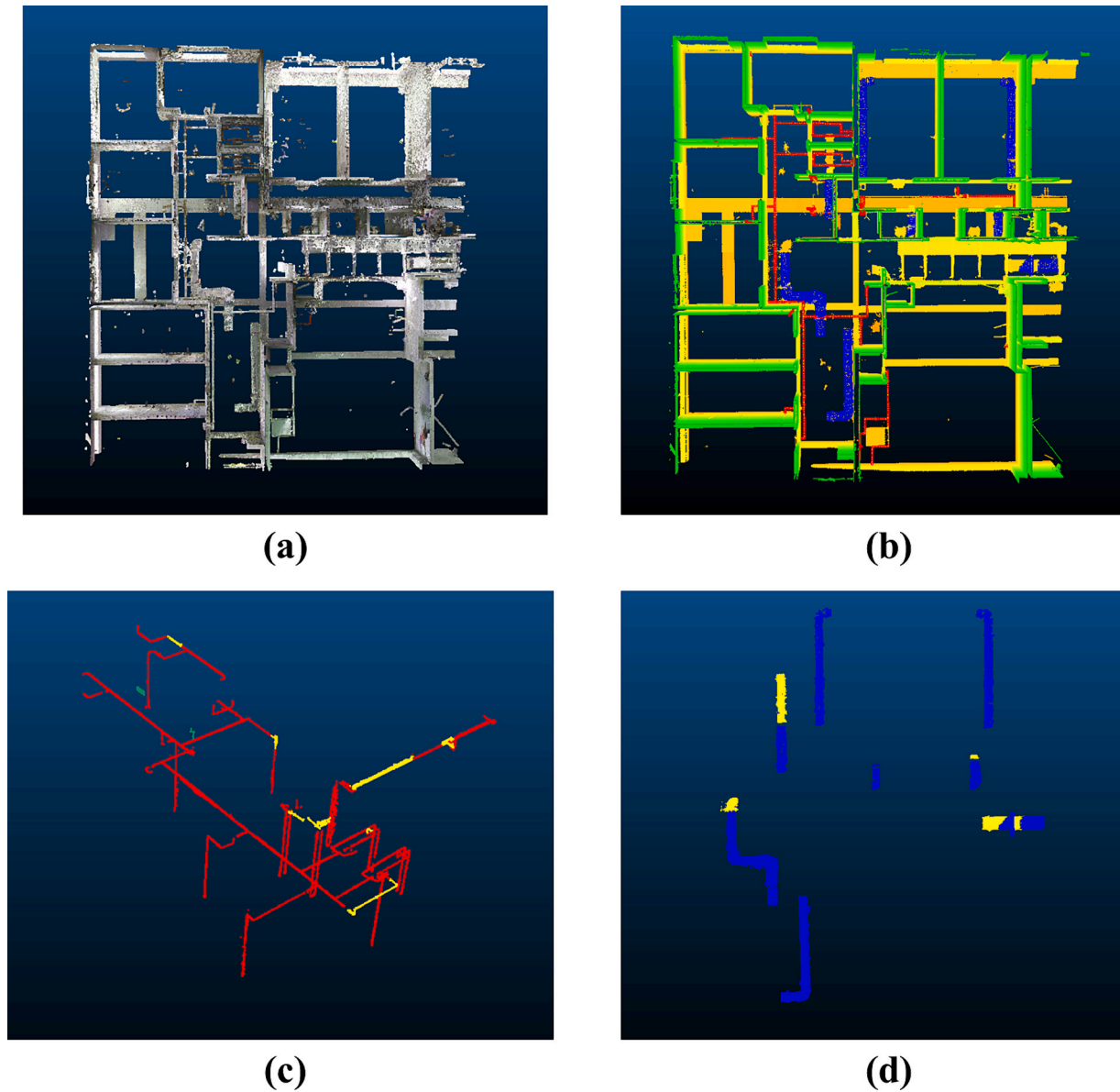


Fig. 12. Point cloud segmentation results from Omni-Scan2BIM: (a) The original point cloud data (floors and ceilings were removed for better visualization); (b) Labeled point cloud data based on Omni-Scan2BIM (slim pipe in red and duct in blue); (c) Segmented point clouds of slim pipes (true positives in red, false negatives in yellow and false positives in cyan); (d) Segmented point clouds of ducts (true positives in blue, false negatives in yellow). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

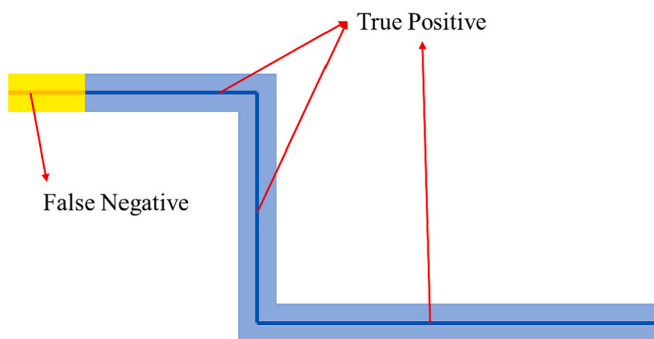


Fig. 13. Metric demonstration: the centerline lengths of the tubular objects were employed to assessing the effectiveness of Scan2BIM methods.

employed LiDAR system could achieve an accuracy of approximately 2 mm and a resolution of 5 mm at a distance of 10 m. The LiDAR had a horizontal field of view of 360° and a vertical field of view of 300°. The laser scanner's internal HDR camera was utilized to capture images with a resolution of 2048 × 2048. Six images were taken for each setup, ensuring full coverage of the scanner's visible area. In total, 29 stations were set to collect point cloud data within the designated area. The point clouds collected from multiple stations were registered using the LEICA CYCLONE software, and the registered point clouds are visualized in Fig. 12(a). The transformation matrices, which describe the relationships between the 2D images and the 3D point clouds, were obtained from the LEICA software. It is important to note that the proposed Omni-Scan2BIM method is not restricted to the specific scanning hardware. The use of LEICA devices in this experiment was solely for demonstration purposes. The experiment specifically focused on two representative tubular MEP components, namely slim pipes (Fig. 9(a)) and ducts (Fig. 9(b)). In this one-shot version of the Omni-Scan2BIM, the sample

Table 1

Comparison of the Scan2BIM results achieved by Omni-Scan2BIM and previous Scan2BIM approaches.

	True Positive (meter)	False Negative (meter)	False Positive (meter)	Recall	Precision
Direct Slicing Method [28]	Cannot work without Manhattan world assumption				
Deep Learning Methods [29,43]	Cannot generalize to unknown scenes				
EdgeWise for Slim Pipe [57]	24.5	58.0	3.1	29.7%	88.8%
Omni- Scan2BIM for Slim Pipe	75.2	7.3	0.8	91.2%	98.9%
Omni- Scan2BIM for Duct	19.4	1.9	0	91.1%	100%

image selection process is based on a small testing. Initially, 6–8 candidate images are chosen based on intuition, selecting images that can effectively represent the general appearance of the target object in the project. These candidate images are then sequentially tested to determine their success in recognizing the target objects in other images. The candidate image that exhibits the best performance, accurately reflecting the general appearance of the target object, is ultimately selected as the final sample.

4.2. Results of validation experiments

In this experiment, the operator annotated the sample image as shown in Fig. 9(a)(b) using the widely recognized labeling tool ‘LabelMe’, to indicate the visual representations of the target components to the developed Omni-Scan2BIM system. Subsequently, the HDR images captured by the laser scanner underwent several processing steps, including deep feature extraction, similarity map generation, prior point sampling, and sequential segmentation. Afterwards, the labels associated with the 2D masks were propagated to the corresponding 3D point clouds, and the 3D shapes of the segmented objects were examined to eliminate potential outliers. Additionally, the labeled point clouds obtained from different setups were fused together using the proximity principle. Finally, considering the spatial relationship between different segments of components, the isolated components which are far away from any other components in the same system are

removed as false positives.

The similarity maps and the predicted masks generated by Omni-Scan2BIM for slim pipes and ducts are presented in Fig. 10 and Fig. 11, respectively. In the similarity maps, the jet colormap was employed, where red indicates high similarity and blue represents low similarity. A higher similarity value suggests a higher probability of the presence of the target objects. It is evident that the similarity maps, generated through feature comparison, successfully indicated the approximate locations of the desired components on the 2D images. This demonstrates the ability of the vision foundation model to extract comprehensive features from on-site images without the need for additional training. Moreover, with the assistance of SAM, the proposed approach accurately segmented the target objects, producing clear outlines on the 2D images based on the sampled prior points.

Upon propagating the 2D labels into the 3D domain and applying label fusion, the target systems, namely the slim piping system and duct system, were successfully segmented from the noisy point cloud data, as illustrated in Fig. 12(b). For clearer visualization of the segmentation results, the 3D point clouds of the slim piping system and the duct system were visualized separately in Fig. 12(c) and Fig. 12(d), respectively. The observations indicated that the vast majority of pipelines, including slim pipes and ducts, were successfully segmented with minimal false positives. In fact, no false positives were observed for ducts, which could be attributed to their larger sizes, facilitating easier recognition. Since the focus of this research lies on tubular components such as pipes and ducts, rather than conducting statistical analysis of point cloud labels, a more meaningful measure is the quantification of centerline lengths covered by the labeled point clouds, as demonstrated in Fig. 13. The quantitative results have been summarized in Table 1, in which the true positive, false negative and false positive were measured by the centerline lengths of the tubular objects in meters. It is evident from the table that 75.2 m of slim pipes were accurately identified, with only 7.3 m of false negatives and 1.4 m of false positives. The recall and precision rates for slim pipes reached 91.2% and 98.2%, respectively. Furthermore, 19.4 m of ducts, out of a total length of 21.3 m, were successfully recognized without any false positives. The recall and precision rates for ducts were 91.1% and 100%, respectively. Based on these findings, it can be concluded that, in the testing scenario, over 90% of tubular components, in terms of length, were successfully recognized and could be further modeled.

In contrast, previous Scan2BIM methods based on deep learning [29,43] are limited in their ability to generalize to unknown scenes without retraining or fine-tuning on purpose-built datasets. Traditional methods that encompass ducts and slim pipes [28,29] only function effectively under the assumption of a Manhattan world, while diagonal

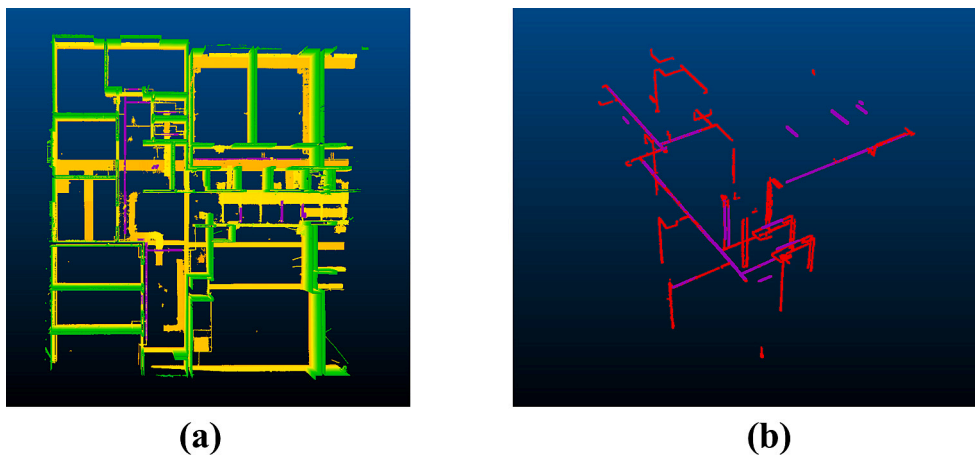


Fig. 14. Point cloud segmentation results from EdgeWise: (a) Labeled point cloud data based on EdgeWise (slim pipe in purple); (b) Comparison of the segmented slim pipes obtained from Omni-Scan2BIM (red) and EdgeWise (purple). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

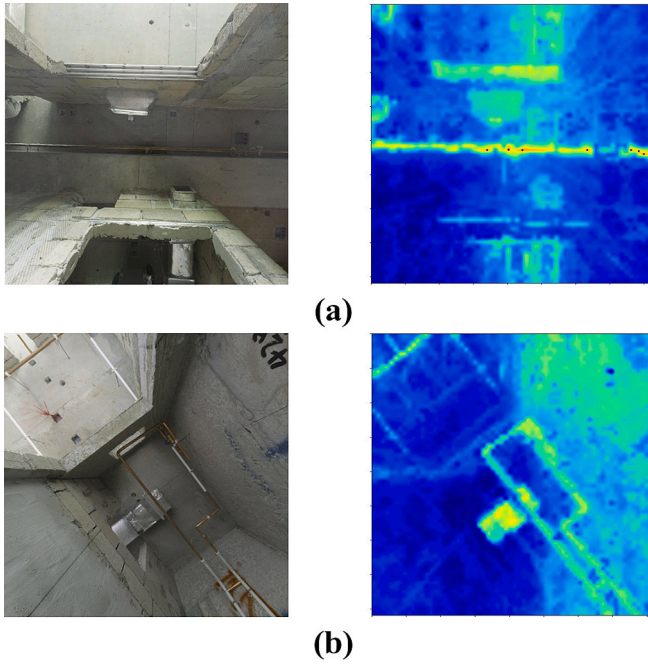


Fig. 15. Failure case analyses: (a) Failure case for the slim pipe mask prediction. The lefthand side is the raw image and the righthand side is the similarity map for slim pipes; (b) Failure case for the duct mask prediction. The lefthand side is the raw image and the righthand side is the similarity map for ducts.

components present significant challenges for these techniques. Commercial software such as EdgeWise [57] can handle circular pipes in novel scenarios through improved geometric methods, potentially utilizing techniques such as Hough Transform or efficient RANSAC. The results of EdgeWise on the tested scenes are also provided in Table 1 and visualized in Fig. 14(a). However, due to the small diameter of the slim pipes, the software exhibits poor performance, with a recall rate of around 30%, significantly trailing behind the performance of the proposed approach, as depicted in Fig. 14(b).

4.3. Failure case analysis

Observing Table 1, it is apparent that there were still some missing

parts (false negatives) in the segmented point clouds, as depicted in Fig. 12(c) and Fig. 12(d). The underlying reason for this phenomenon, according to the mechanism of the proposed Omni-Scan2BIM method, was the failure to recognize the target objects in the image from the nearest setup. Fig. 15(a) and Fig. 15(b) illustrate two examples of such failures, one involving slim pipes and the other involving ducts, respectively. It can be observed that the slim pipes in Fig. 15(a) and the ducts in Fig. 15(b) were not recognized due to the low similarity in appearance between the objects in the captured on-site images and the objects in the sample images.

The slim pipes depicted in the sample image shown in Fig. 9(a) are predominantly covered by white plastic insulation material, whereas the slim pipes in the captured image illustrated in Fig. 15(a) are primarily composed of bronze metal. This significant variation in appearance contributes to the low similarity observed between the target object in the captured image and the corresponding object in the sample image.

In the case of ducts, the sample image displayed in Fig. 9(b) was acquired in an environment with abundant lighting conditions. Conversely, the captured images depicted in Fig. 15(b) were obtained in a confined and dimly lit MEP room. Consequently, the ducts could not be recognized due to the insufficient illumination levels present in these images.

To address this limitation, a potential solution would involve incorporating multiple sample images that encompass a wider range of possible situations. This approach would enable the system to learn from various appearances and adapt to different scenarios. The authors are actively working on the development of the few-shot version of Omni-Scan2BIM to enhance its robustness under different lighting and material conditions.

Additionally, the recognition performance of the Scan2BIM process can be significantly influenced by the quality of captured images and the level of occlusion present in highly dense MEP components. It is evident that recognition outcomes are adversely affected by very poor-quality images, whereas higher-quality images produce more reliable results. Nevertheless, establishing a quantitative threshold for image resolution which defines a minimum requirement may vary depending on the specific objects being analyzed, thereby requiring further research. To address the challenges posed by occlusions, one potential solution is the deduction approach, wherein closely arranged MEP components are collectively treated as a set and deducted as a whole. If a portion of the set has already been successfully recognized, the entire set is presumed to be complete based on the information provided by the as-designed BIM. This method effectively utilizes the knowledge of the overall

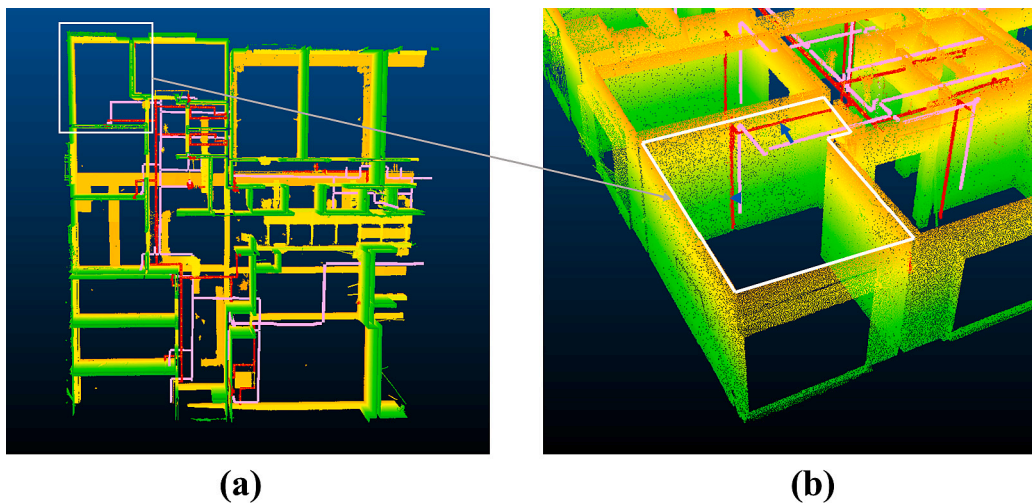


Fig. 16. Potential applications of Omni-Scan2BIM by integrating Scan-vs-BIM: (a) As-built slim pipes in red versus as-designed slim pipes in pink (b) A zoom in view of the comparison between as-built pipes and as-designed pipes for one room. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

system design to infer the presence and completeness of obscured components.

4.4. Potential applications

The proposed Omni-Scan2BIM methodology holds significant potential in facilitating the updating of as-designed models, monitoring construction progress, and conducting engineering quality inspections for MEP systems. As highlighted in the introduction, the presence of assembly deviations and frequent design alterations leads to substantial disparities between the actual as-built conditions and the corresponding as-designed BIM representations. Consequently, updating as-designed models, monitoring construction progress, and conducting quality inspections pose particular challenges for MEP systems, distinguishing them from architecture and structural systems.

While Omni-Scan2BIM has demonstrated promising results in recognizing MEP systems and generating parametric as-built models, there still exists a significant lack of data and details pertaining to the geometric model. This includes globally unique identifiers (GUIDs), supplier information, insulation conditions, tags, and other relevant attributes. As this kind of information is typically entered manually by BIM modelers into the as-designed BIM, it is more prudent to update the as-designed BIM to the as-built positions by establishing correspondences between the two data sources, rather than re-entering the information manually. In the conducted experiment, the as-built conditions of slim pipes are represented in red, while the corresponding as-designed model is depicted in pink, as illustrated in Fig. 16(a). While the overall trends of the as-built and as-designed pipes remain consistent, deviations are observed in specific locations. A more detailed illustration of this argument is presented in Fig. 16(b), which provides a zoomed-in view of a specific room. By identifying the corresponding objects in the as-built model, the as-designed model can be automatically updated to reflect the as-built positions along with comprehensive information, as indicated by the blue arrows in Fig. 14(b). In addition to efficiently updating the as-designed model, this correspondence between the as-built and as-designed models can also aid in construction progress tracking and quality inspection in real projects. The logic behind this approach is as follows: for each object in the as-designed model, if the corresponding objects can be found in the as-built model, the construction status of the object would be marked accordingly.

5. Conclusions

This study introduces Omni-Scan2BIM, an innovative Scan2BIM approach that leverages vision foundation models with exceptional generalizability. Unlike previous methods, Omni-Scan2BIM can be readily applied to unseen scenarios without the need for training or fine-tuning. In the proposed method, users are required to annotate the target object on a single sample image. Visual features are then computed using an all-purpose feature extraction model, such as DINOv2, for the specified object in the sample image as well as hundreds of additional on-site images. By comparing these features, similarity maps are generated for the captured on-site images, indicating the potential likelihood of each pixel belonging to the target component. Subsequently, prior points are sampled from the local maxima of the similarity maps and fed into the generalized segmentation model, SAM, for precise boundary segmentation. By leveraging the transformation matrix between 2D images and 3D point clouds, the 2D labels are propagated to the corresponding 3D point cloud data. Furthermore, a joint shape analysis involving both 2D and 3D data is performed to alleviate potential outliers in the segmentation process. To ensure accuracy and completeness, labels from different setups are fused based on the proximity principle. This fusion process enhances the consistency and reliability of the final segmentation results.

The proposed technique was validated using data collected from an actual construction site in Hong Kong. In the case of slim pipelines,

Omni-Scan2BIM demonstrated successful reconstruction over a length of 75.2 m, within a total range of 82.5 m. The achieved recall and precision values were 91.2% and 98.9% respectively. In comparison, the state-of-the-art commercial software EdgeWise, which relies on traditional geometry analysis, only reconstructed 24.5 m of the pipelines, with a recall of 29.7% and precision of 88.8%. Regarding ducts, Omni-Scan2BIM achieved successful reconstruction over a length of 19.4 m, within a total range of 21.3 m. The recall and precision values for this category were also reported as 91.1% and 100.0% respectively. It is worth noting that there is currently a scarcity of mature automatic solutions for rectangular duct BIM reconstruction, both in commercial markets and research domains. Traditional deep learning models also face challenges in direct generalization to new sites without retraining or fine-tuning on custom-designed datasets. Due to these inherent limitations, the results obtained from these methods cannot be easily replicated and are therefore not presented in this study.

While this study has yielded promising results, it is important to acknowledge certain limitations. The developed Omni-Scan2BIM technique relies on a single sample image, which may not adequately capture the variability in appearance or account for illumination variations. Consequently, false positives were observed during the validation phase of this research, highlighting the need to incorporate a slightly larger number of samples to encompass a broader range of scenarios. Furthermore, this study introduces three potential application directions for the developed Omni-Scan2BIM methodology, namely as-designed model updating, construction progress tracking, and engineering quality inspection. However, these directions warrant further exploration and investigation to fully understand their potential and feasibility.

CRediT authorship contribution statement

Boyu Wang: Conceptualization, Data curation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Zhengyi Chen:** Conceptualization, Data curation, Formal analysis, Methodology, Software, Supervision, Writing – original draft, Writing – review & editing. **Mingkai Li:** Conceptualization, Methodology, Project administration, Resources, Supervision, Validation, Visualization. **Qian Wang:** Project administration, Resources, Software, Supervision. **Chao Yin:** Data curation, Methodology, Software, Supervision, Validation. **Jack C.P. Cheng:** Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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